



EZ1290 V2.6

Assembly Manual

By Martin Adriaanse

EZ1290 PCB Introduction

- The EZ1290 project came to be in late 2005 out of a fervent desire to create a high quality microphone preamp for the DIY audio community. The design criteria were that it should stand its ground with the best in the industry - at a fraction of the cost - and that it needed to be relatively easy to build (hence the “EZ” in the product name).
- The EZ1290 is a carefully designed replica of the classic Neve 1073 preamp without EQ and Line-Input. The 1290 number originates from Neve’s very own 1290 preamp which was essentially a 1073 sans EQ. It should be noted however that Neve’s original 1290 was wired slightly differently than a 1073 with EQ disengaged. The EZ1290 is neither a Neve 1290 clone, nor a Neve 1272 with added gain; it operates like a 1073.
- Gain is produced by two or three gain stages that are laid out true to the original BA183AV amplifier:
 - BA183NV – the pre stage
 - BA183AM – the driver stage
 - at 55dB gain and above, a third pre stage (BA183NV) is switched in to supply additional gain.
- The BA183AV board pre-dates the BA283AV board and although the two sound identical, the BA183AV is a little roomier and easier to assemble.
- Besides leaving out EQ and line input, the 80dB gain position was also eliminated in order to facilitate a 12 position, PCB mounted rotary switch. Aside from a reduction in total gain, this omission has no measurable or perceivable effect on sound.
- Through its various revisions, the EZ1290 has had a few capacitor changes based on feedback from a renowned Neve tech. Most notably, the decoupling caps and a few other electrolytic caps at the output stage were increased in value to enhance low frequency response.
- The latest revision, V2.6, includes footprints for radial polypropylene capacitors, making the project even easier to build while maintaining the lovely sound of the original.
- This project would not have been possible, had it not been for the generous help from Joe Malone (JLM Audio), Geoff Tanner (Aurora Audio) and Colin Adshead (Audiomaintenance.com). Rupert Neve also deserves our gratitude for enriching our lives with his excellent designs.

Who should build it

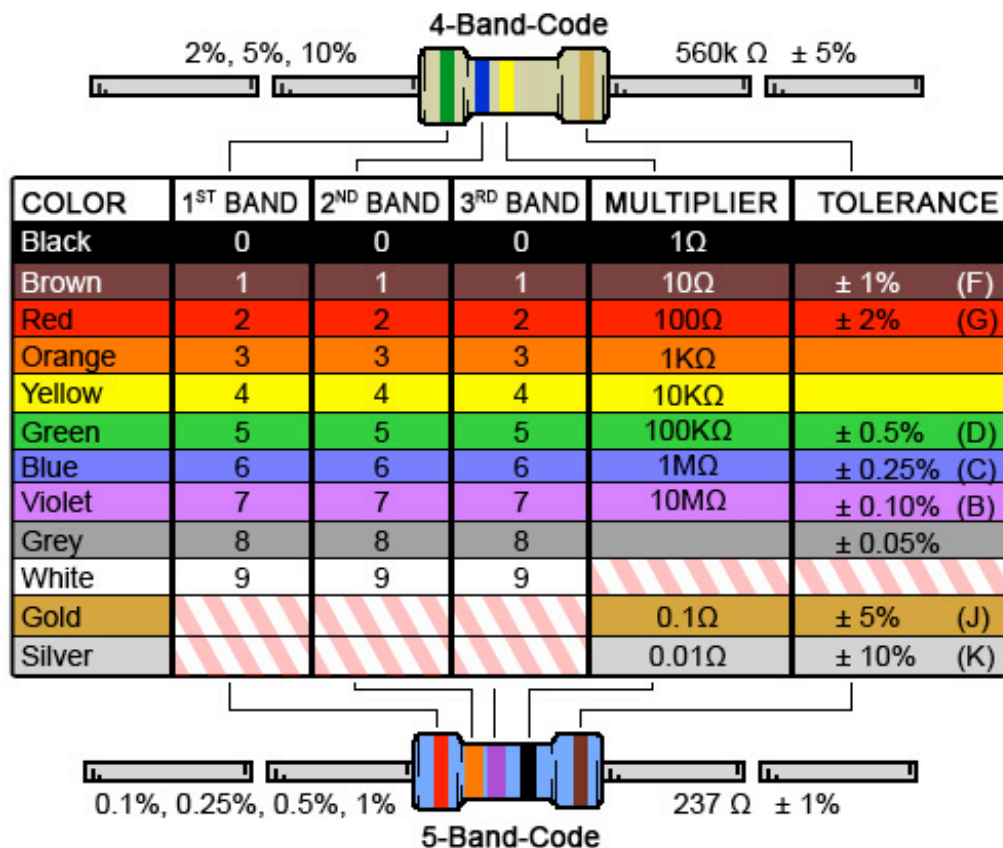
- This project has been built successfully by complete novices with no previous electronics experience and experienced technicians alike.
- It is a fairly easy build if you follow the directions, but you do have to pay careful attention to things like component values and capacitor orientation.
- You do have to know how to use a soldering iron and a volt meter. There are many videos available on the web explaining how to use either.
- Having access to an oscilloscope and signal generator is extremely useful but not required to get the EZ1290 up and running.

Required tools

- A good soldering iron with fresh tip (Weller, Hakko)
- .025" 60/40 Rosin Core solder (Kester 44)
- Wire cutters
- Wire strippers
- Small (needle nose) pliers
- Resistor lead forming tool (1/4-1/2W resistors)
- Desoldering vacuum pump
- A screwdriver set
- A drill and a metal file for prepping the enclosure
- 15/16" Greenlee for punching XLR holes

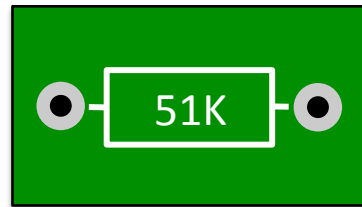
Resistors

- The EZ1290 *Bill of Materials* specifies 1% metal film resistors. They are coded using 5 colored bands and their values can be derived from the chart below. If unsure about value, verify with an Ohm meter.

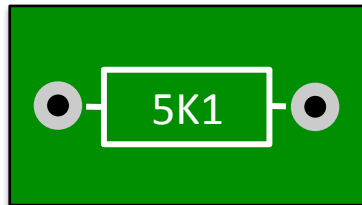


Resistors

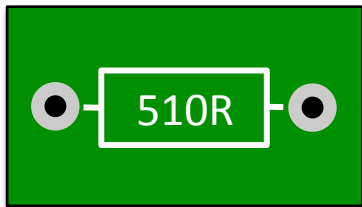
- The first step in building the PCB should be to mount all the resistors. Resistors are symbolized on the PCB using the following format:



Indicates a 51,000 Ohm resistor



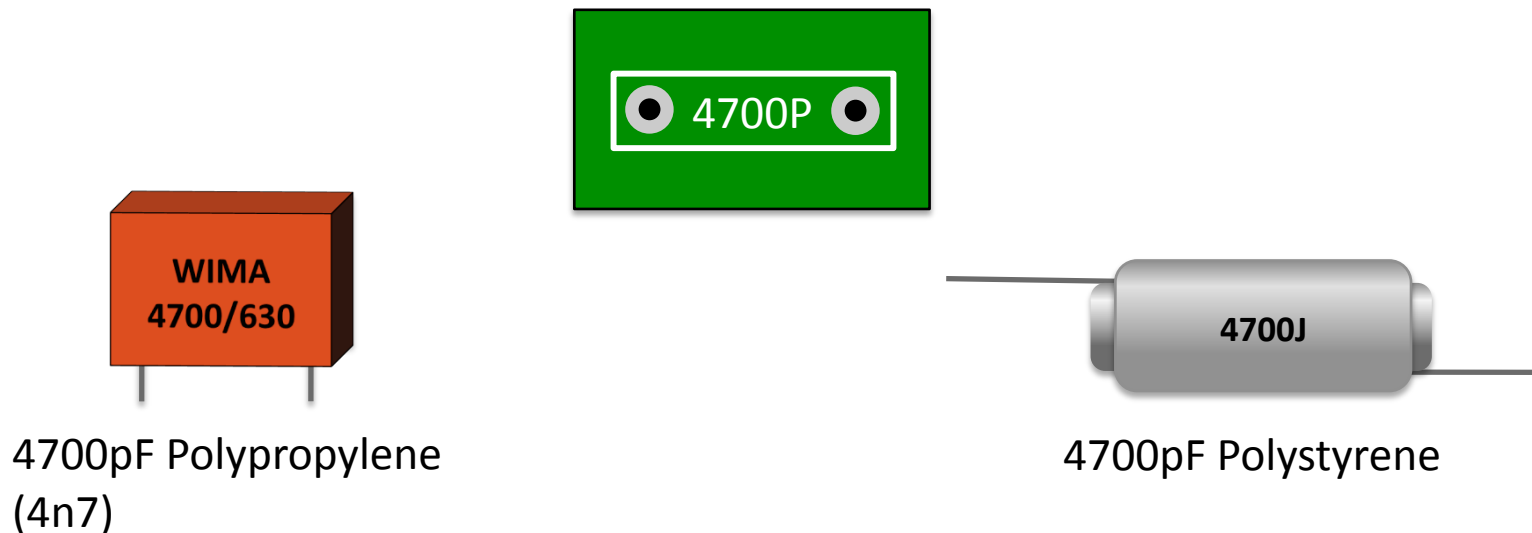
Indicates a 5,100 Ohm resistor



Indicates a 510 Ohm resistor

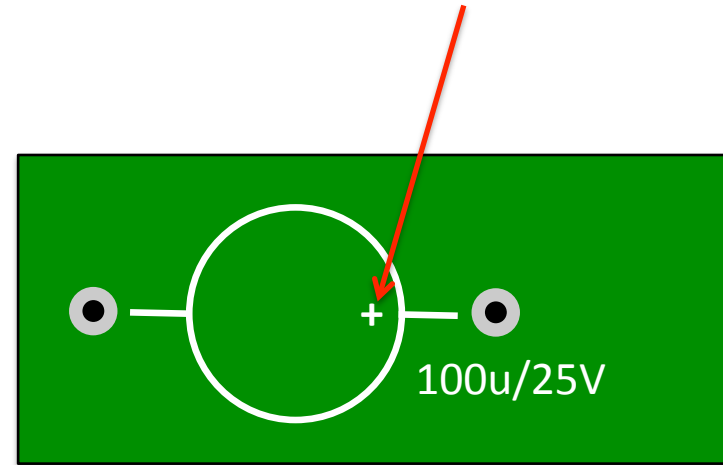
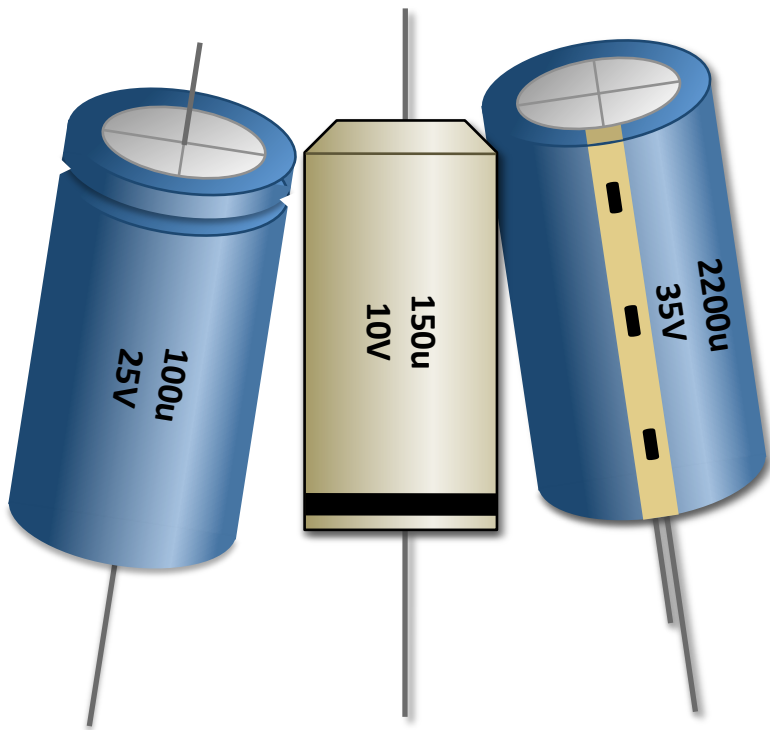
Polypropylene / Polystyrene Capacitors

- After the resistors are soldered to the PCB, the polypropylene capacitors are next. The original 1073 used polystyrene capacitors, but they are getting harder to find and can be destroyed by excessive soldering heat. Polystyrene caps can certainly be used on the EZ1290 board as before, but I now recommend polypropylene. Polypropylene capacitors are not sensitive to polarity and are symbolized on the PCB as follows:



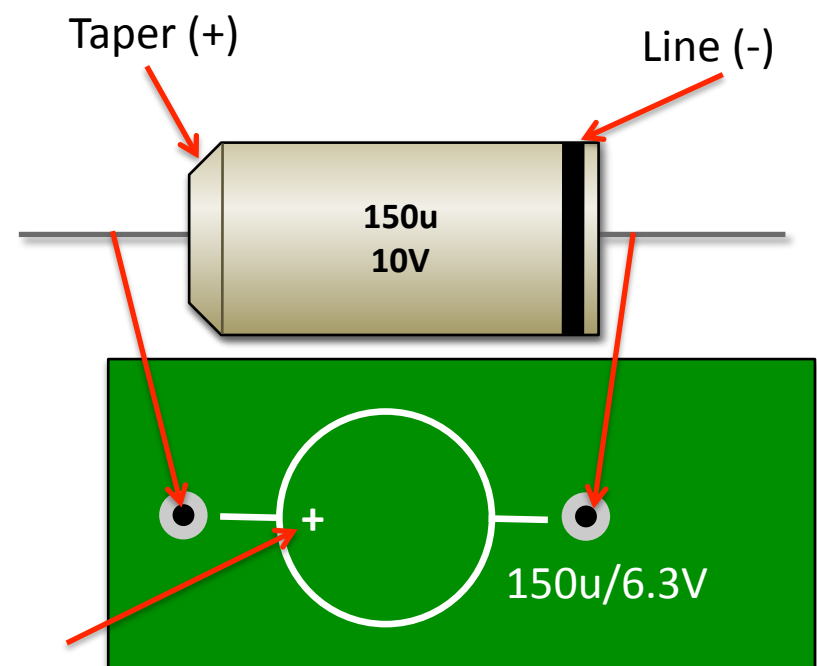
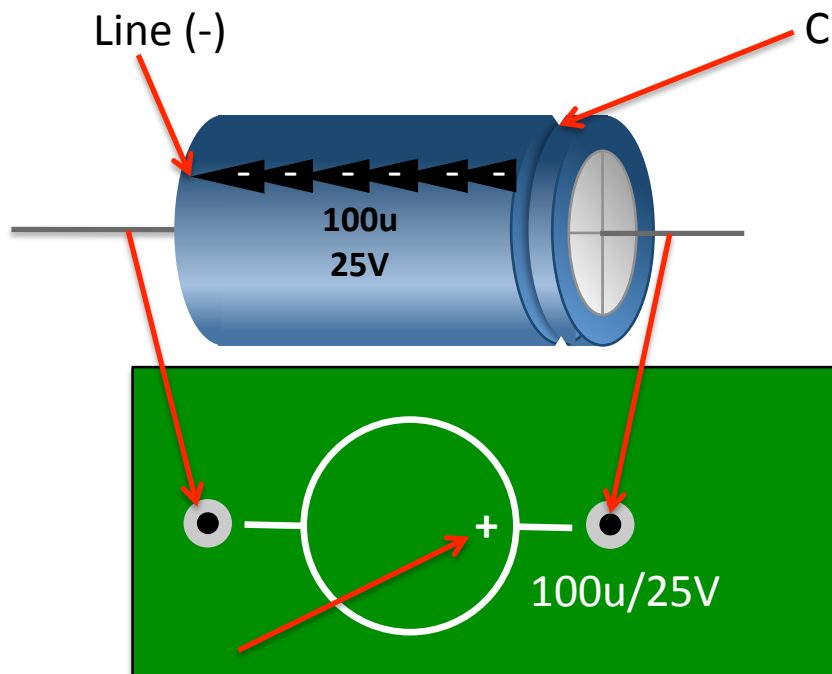
Electrolytic Capacitors

- After the polypropylene capacitors are soldered to the board, the electrolytic capacitors are next. Careful attention needs to be given to the proper orientation of the capacitor on the PCB as they are polarity sensitive. Electrolytic capacitors are symbolized on the PCB as follows: (Note the polarity (+) sign)



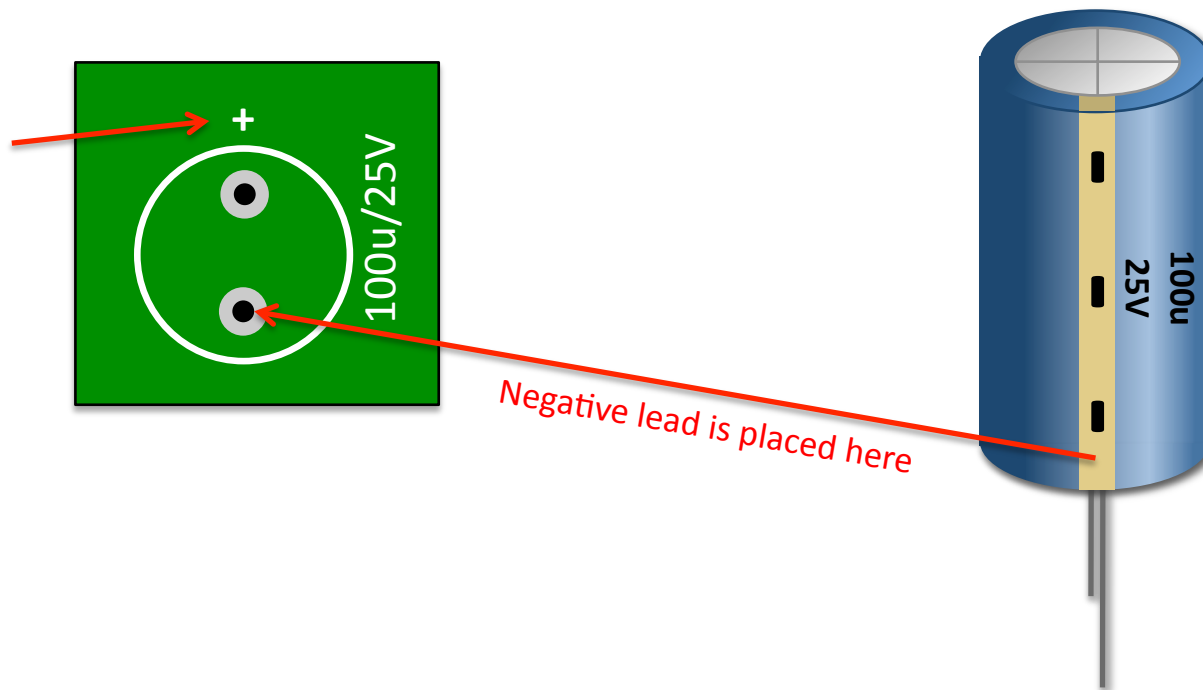
Electrolytic Capacitors

- Axial electrolytic capacitors usually have a crimp or taper indicating the positive end, and a line indicating the negative end. They are symbolized and should be placed as follows:



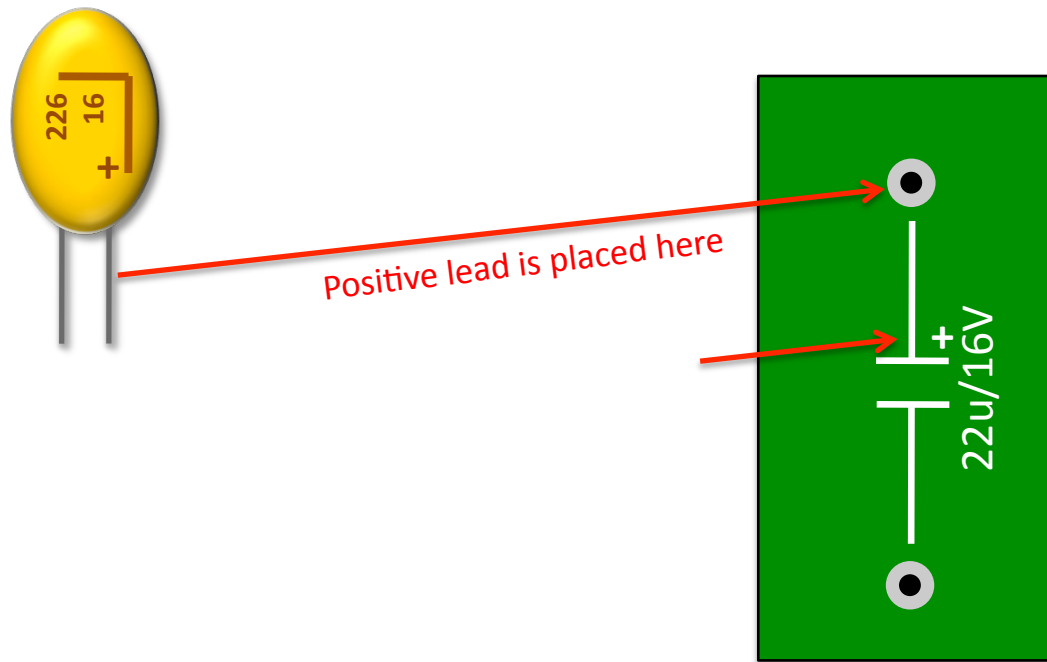
Electrolytic Capacitors

- Radial electrolytic capacitors have a line indicating the negative lead side. They are symbolized and should be placed on the PCB as follows:



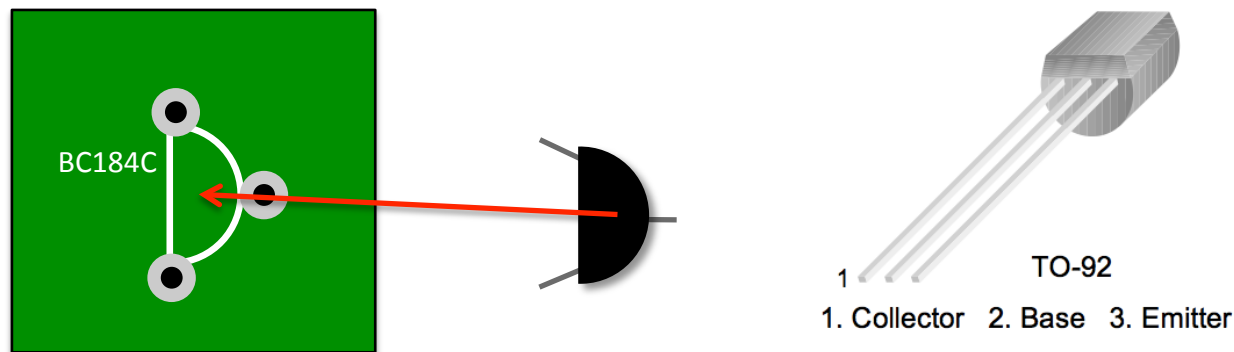
Tantalum Capacitors

- Tantalum capacitors have a line and + sign indicating the positive lead side. The leads should be bent out to align them with the through holes on the PCB. They are symbolized and should be placed as follows:



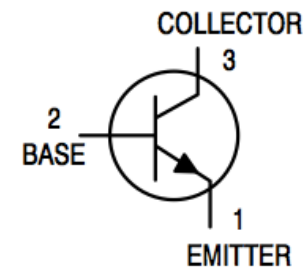
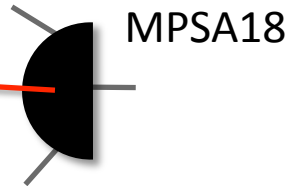
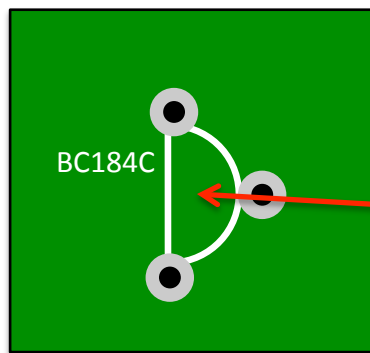
Transistors

- Transistor BC184C, though currently obsolete, can easily be sourced from ebay. It is symbolized on the PCB as pictured below and its D-shaped body should be lined up with the D-shaped silkscreen on the PCB.

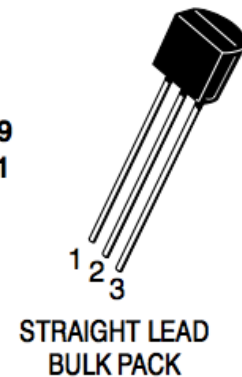


Transistors

- Various substitutes may be used in place of transistor BC184C (see BOM). A popular substitute is transistor **MPSA18**. Special precaution should be taken when using this transistor as the pinout is different. It should be rotated 180 degrees on axis compared to BC184C.

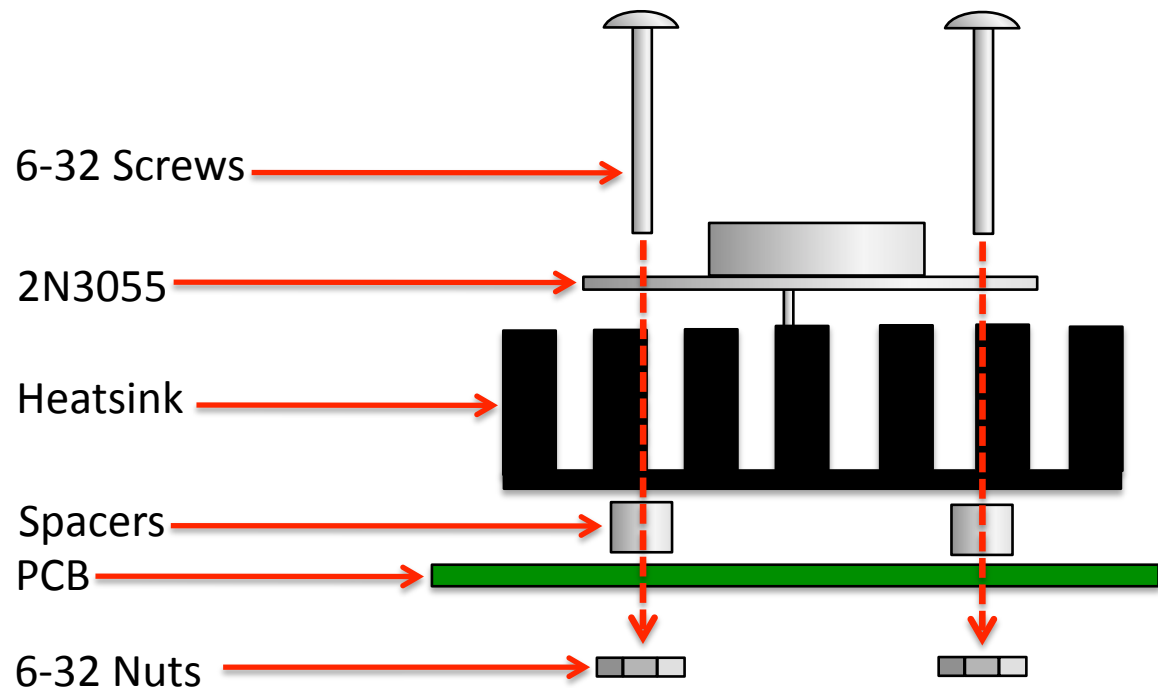
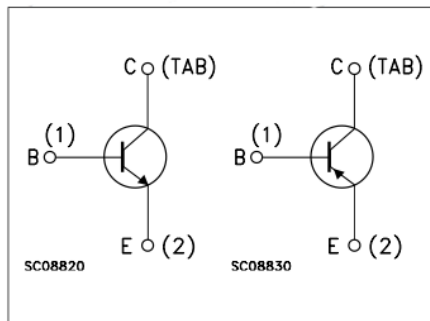
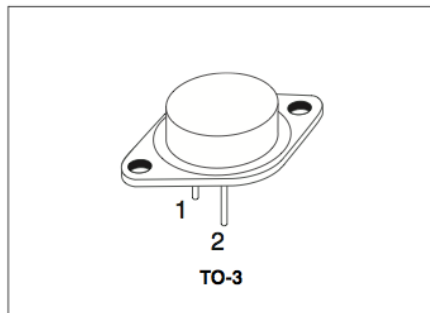


TO-92
CASE 29
STYLE 1



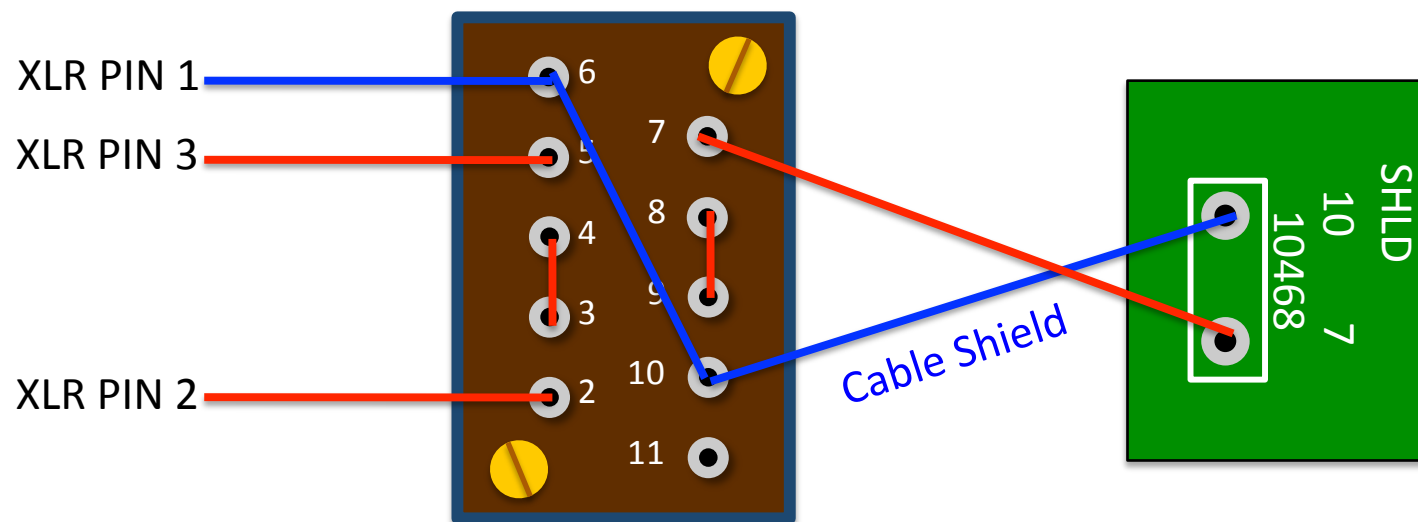
Transistors

- 2N3055 is bolted and soldered to the PCB as pictured below.
- Bolt 2N3055 to the PCB **before** soldering to avoid putting strain on the solder pads.
- Spacers should be about 3/16" high



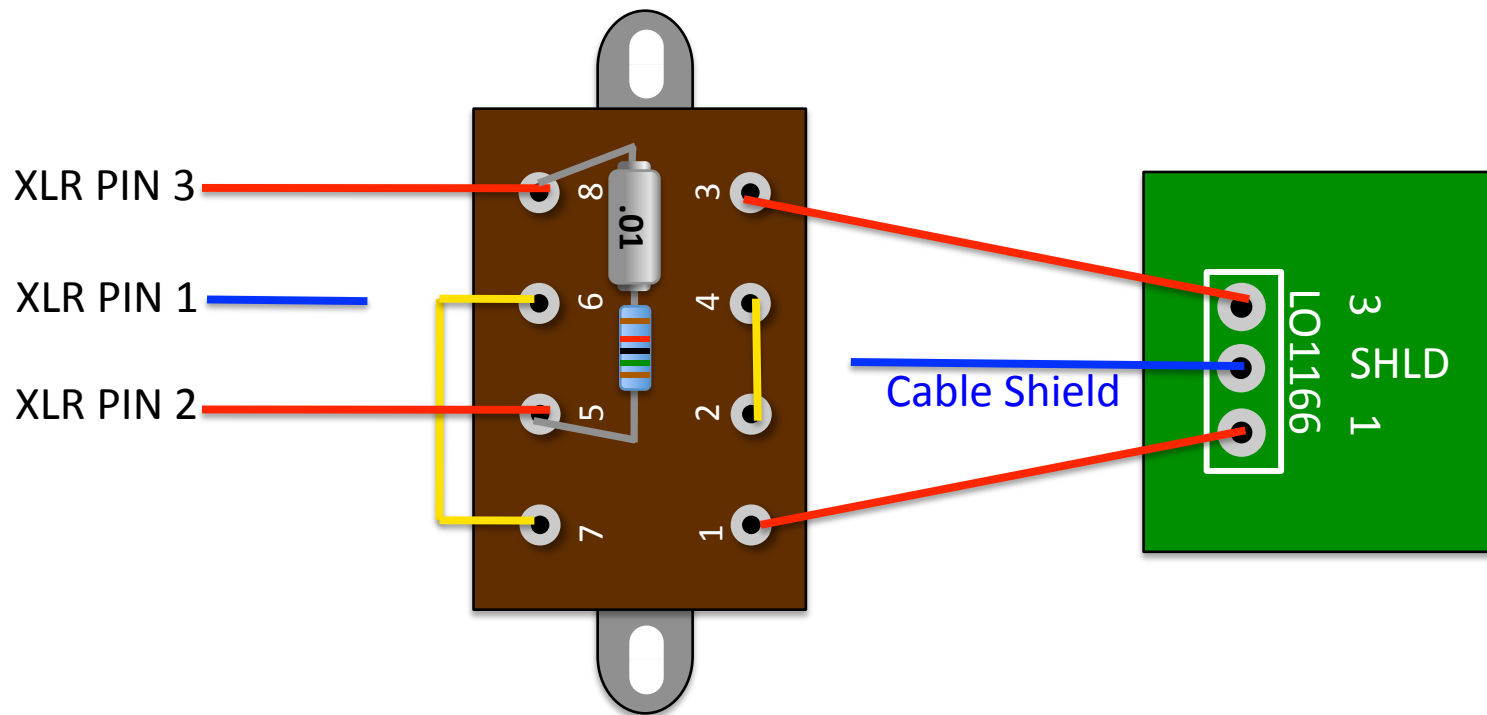
Transformers

- A Carnhill VTB9045 (10468) input transformer is used for the EZ1290. It's connected to the PCB and the input XLR connector as shown below.
- You'll want to use shielded wire (Canare L-2B2AT) from the PCB to the transformer and from the transformer to the XLR connector.
- Wire PCB → Pin 10 and Pin 6 → XLR Pin 1 through the “drain wire” (shield).



Transformers

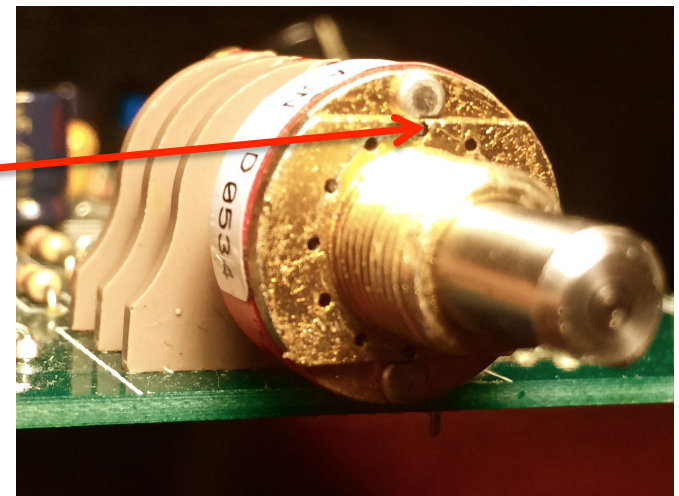
- A Carnhill VTB9049 (LO1166) output transformer is used for the EZ1290. It's connected to the PCB and the input XLR connector as shown below. You'll want to use shielded wire (Canare L-2B2AT) from the PCB to the transformer, and from the transformer to the XLR connector.
- Please note the .01uF cap and 1K5 resistor connected between Pin 8 and 5.
- Use XLR PIN1 for cable shield only; leave disconnected otherwise to prevent ground loops.



Rotary Switch

- The rotary switch selects the gain setting of the preamp. A Grayhill 71BDF30-03-1-AJN switch is used for this purpose.
- Use some fine .015" solder and a sharp soldering tip. The switch solder pins are only .075" apart and it's easy to get a solder bridge between them.
- Some of the pins on the middle row don't have solder pads; they are unused switch positions. Don't add solder to these.
- Insert one stop pin at the 12o'clock position. This will give you a stop between 75dB gain and 20dB gain. Use the silver sticker to cover the stop pin to prevent it from falling out.

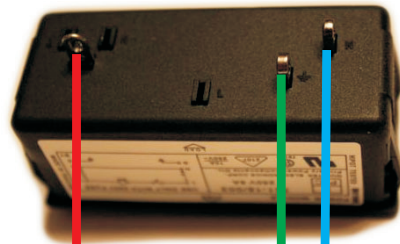
Stop Pin



JLM POWER PLANT Power Supply

**QUALTEK
761-18/003**

2.6" X 1.063" Panel Cutout



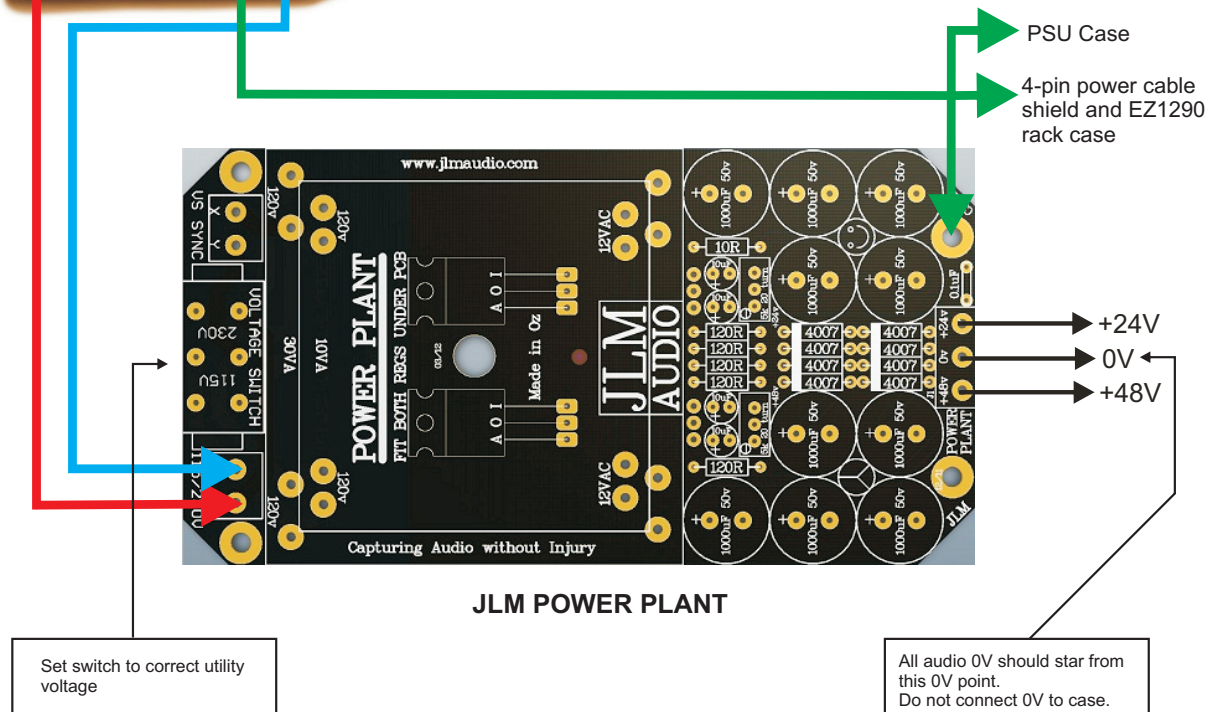
The JLM Power Plant can power up to 4 EZ1290 modules.

Fuses:

Use a 250mA slow blow fuse for 230v and a 500mA slow blow for 115v.

Make sure all voltages are verified/adjusted before connecting to EZ1290.

Connect/test one EZ1290 at a time initially.



PSU Case

4-pin power cable
shield and EZ1290
rack case

+24V

0V

+48V

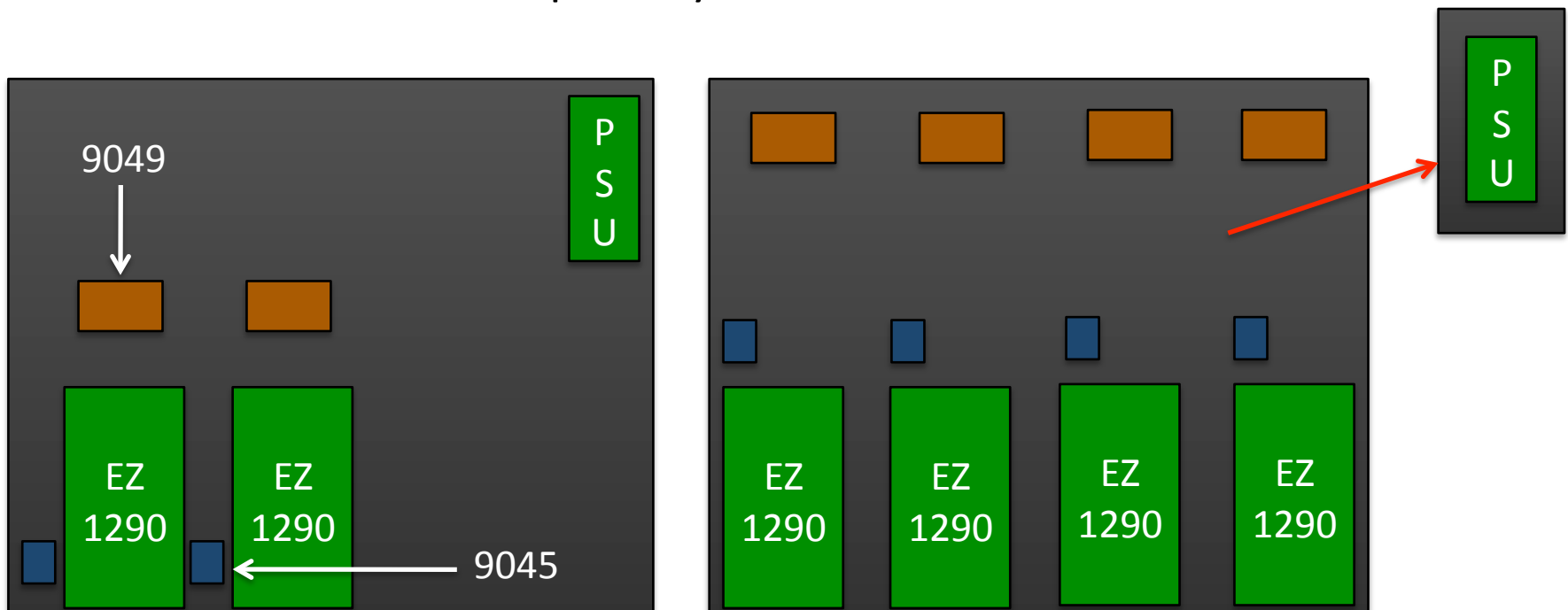
JLM POWER PLANT

Set switch to correct utility
voltage

All audio 0V should star from
this 0V point.
Do not connect 0V to case.
If your grounding is correct
you should measure 10ohm
between rack case and 0V.

Layout Considerations

- Power supply should be kept away from audio as much as possible to avoid EMI (hum).
- Two channels may be mounted into a 2U enclosure with the EZ1290's on the far left, power supply on the far right.
- Four channels may be mounted into a 2U enclosure with an external power supply.
- VTB9045 should be kept away from VTB9049.

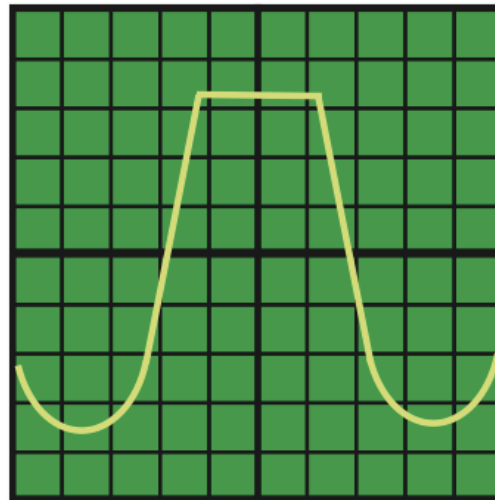
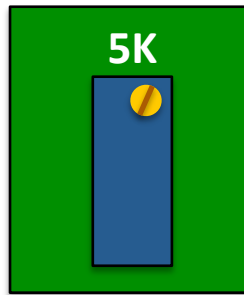


Bias

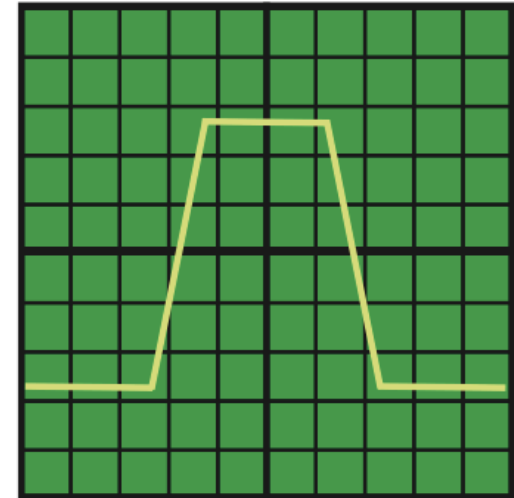
- After everything's been assembled, you'll need to set the bias of the driver stage. An oscilloscope and signal generator is required to do this properly.
- Set the gain switch to full gain (75dB) and then slowly increase the output of the signal generator – set to 1K sine wave - into the mic input until clipping is indicated on the oscilloscope.
 - It's ok if you actually hear the mic pre at this point
- More than likely the waveform is only clipping on one side.
- Adjust the 5K trimpot until clipping is symmetrical on both sides of the waveform.
- To roughly set bias in the absence of having proper test equipment, adjust 5K trimpot until TR3 collector (2N3055 casing) measures $\sim 22.66V$.

Bias

Adjust 5K trimpot until clipping is symmetrical on both sides of the waveform.



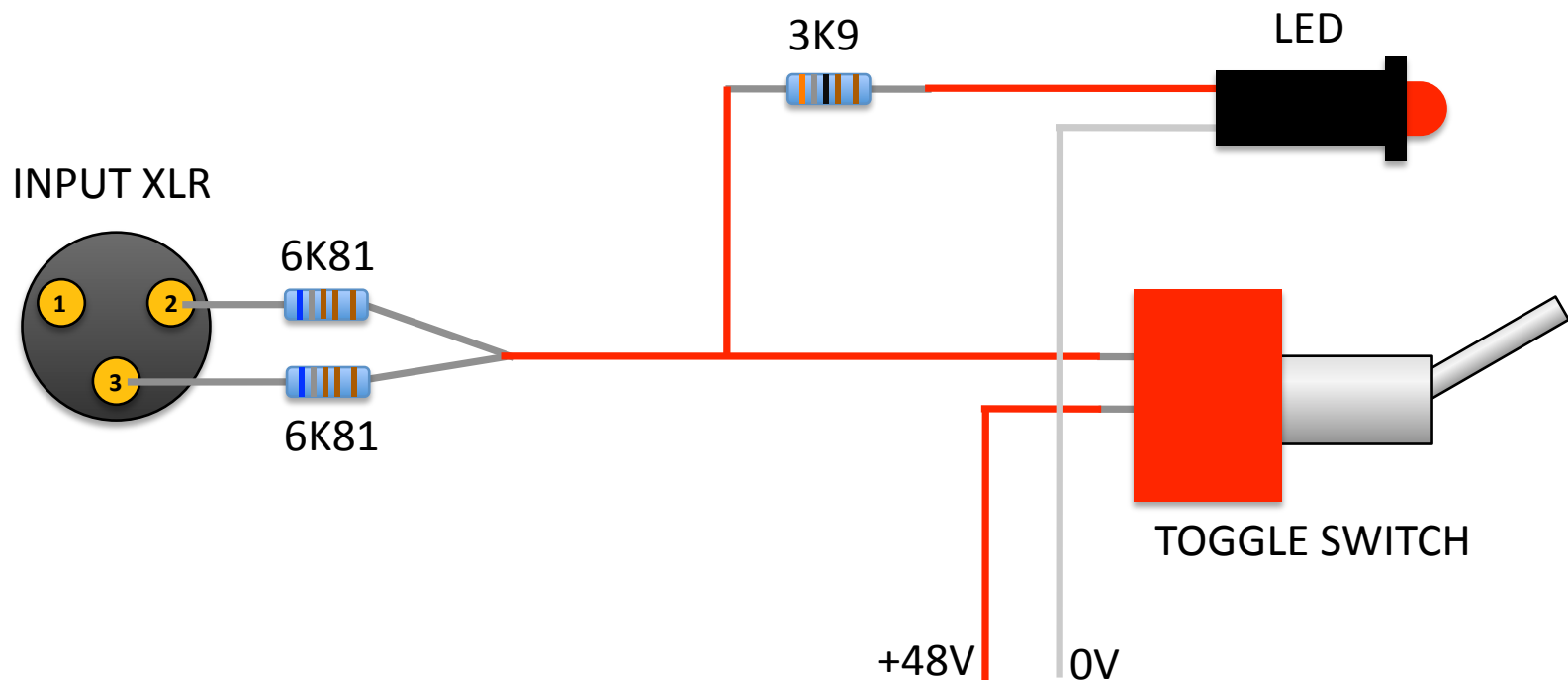
Asymmetrical Clipping



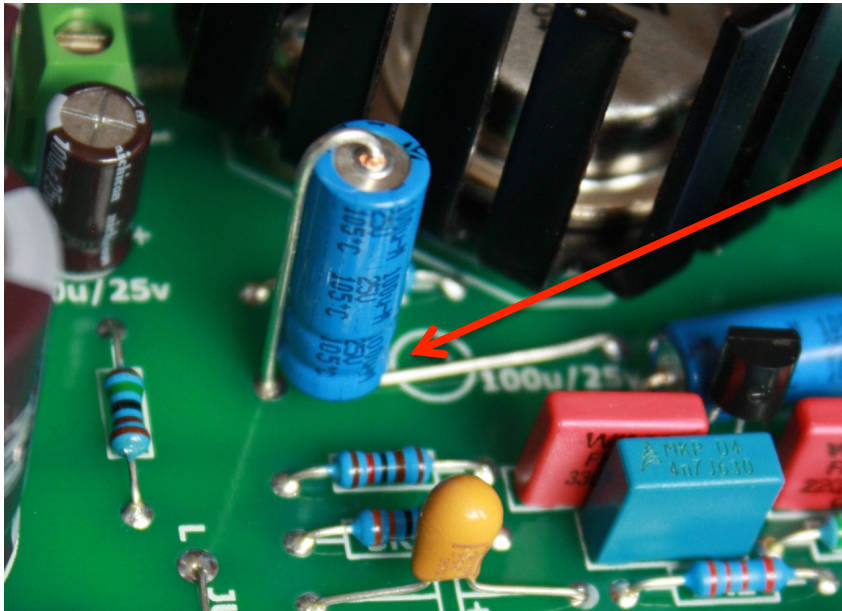
Symmetrical Clipping

Phantom Power (48V)

- Phantom power is connected as depicted below. It's important that the 6K81 resistors are closely matched in value. Buy a few extra and pick out the two closest matching values with an Ohm meter. I recommend wrapping the resistors in 1/8" heatshrink

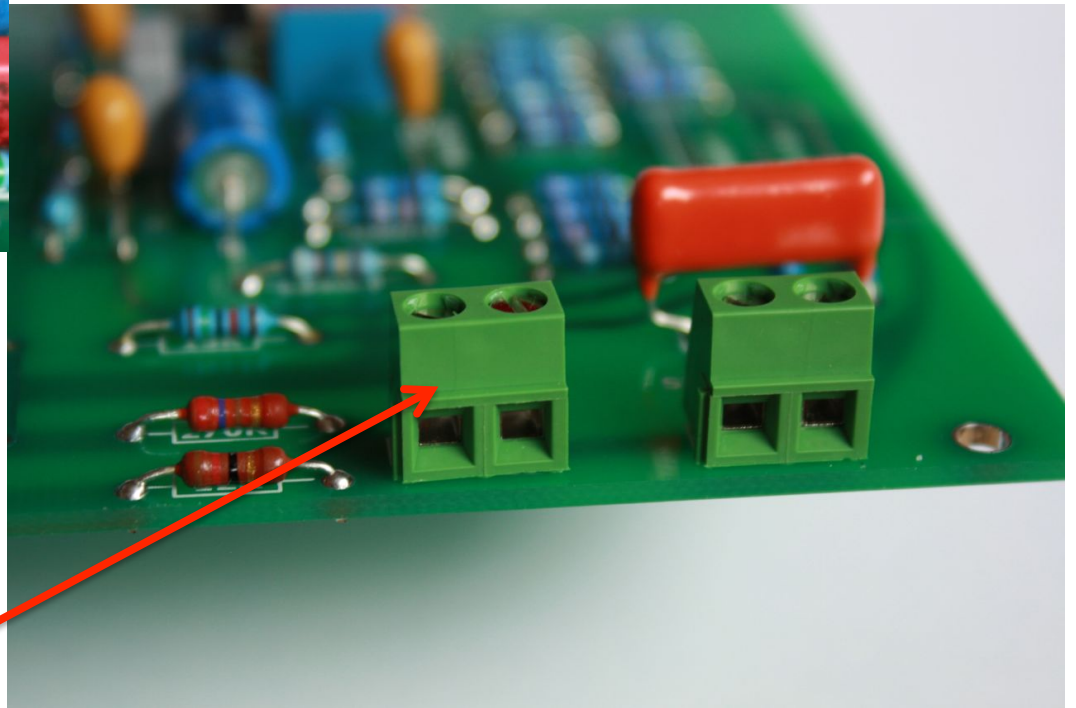


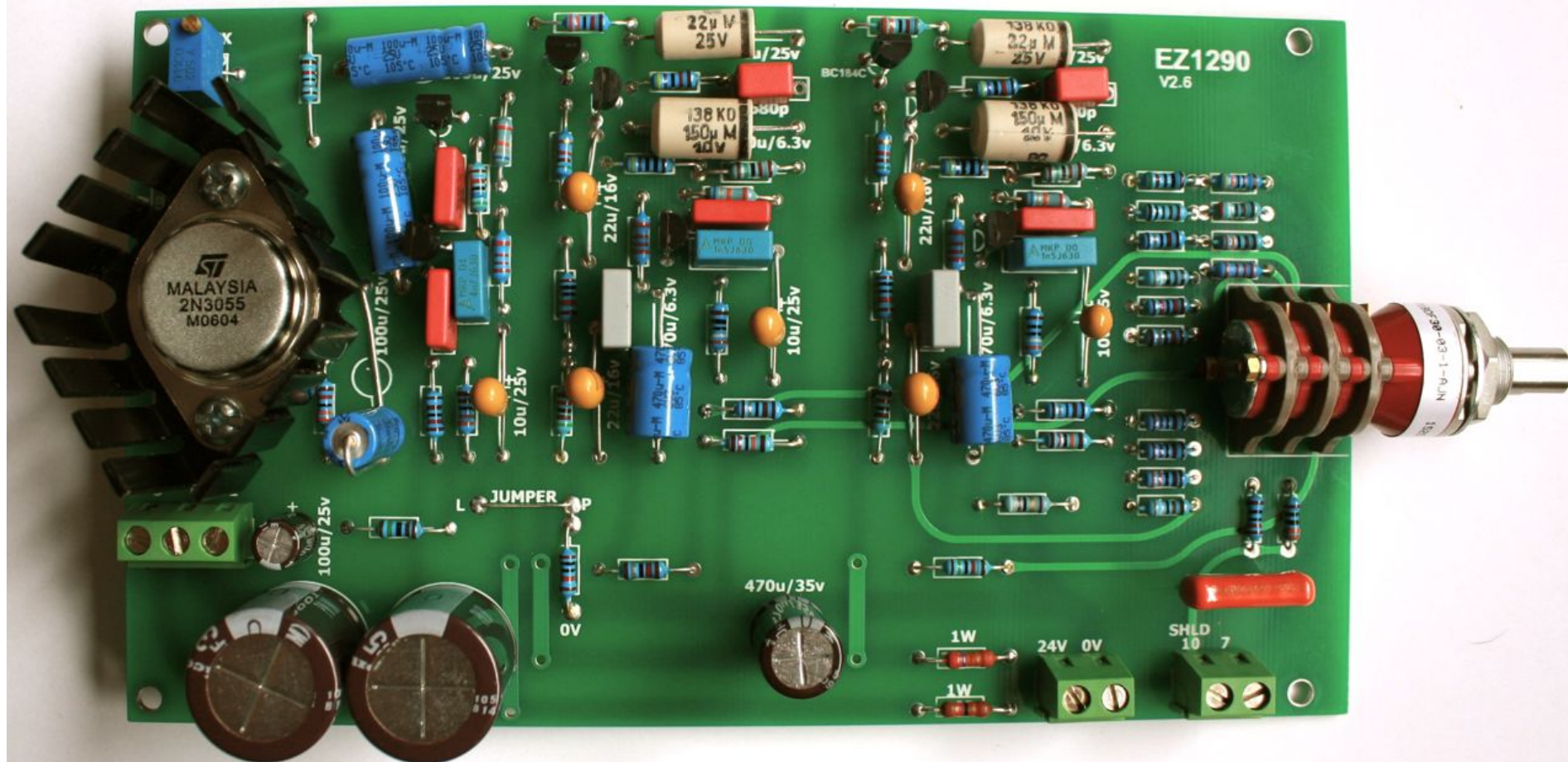
Other Considerations

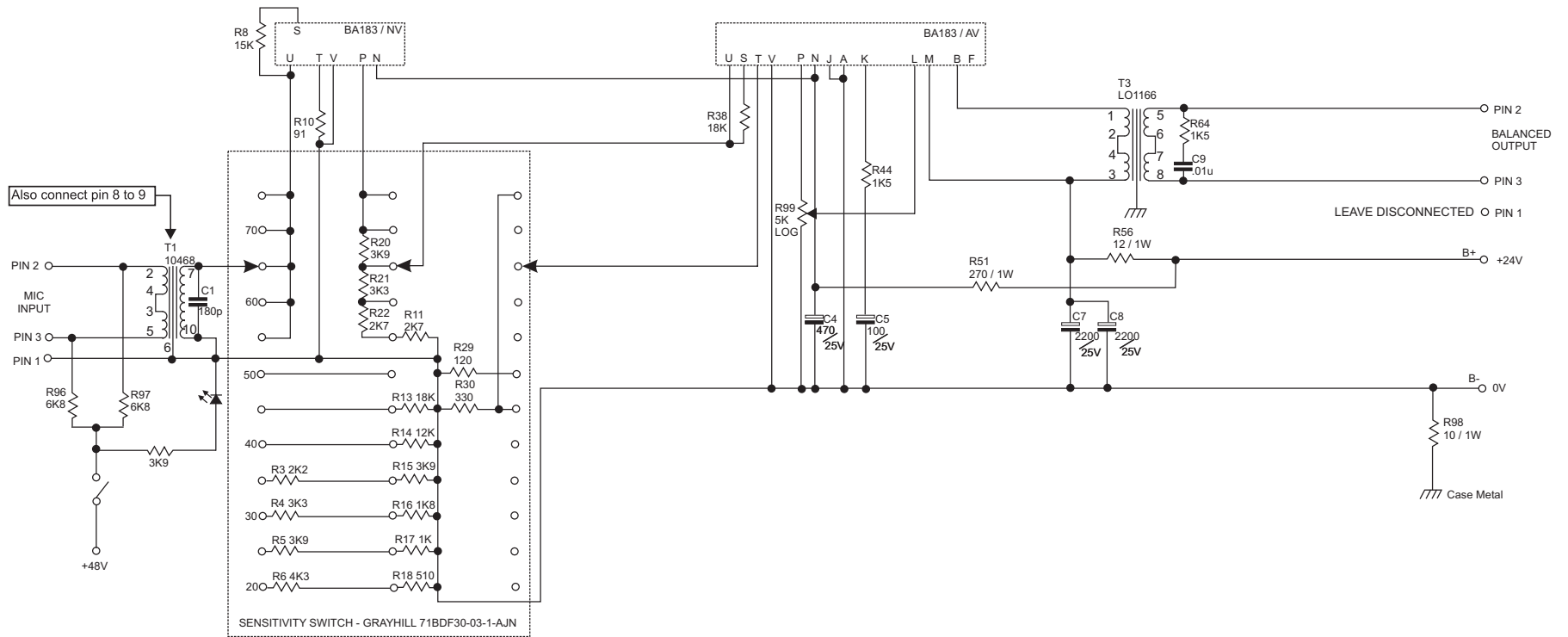


This 100uF/25V axial cap should be mounted upright to clear 2N3055 heatsink

Terminal blocks may be used for all wired connections, however, it is best to solder all connections.

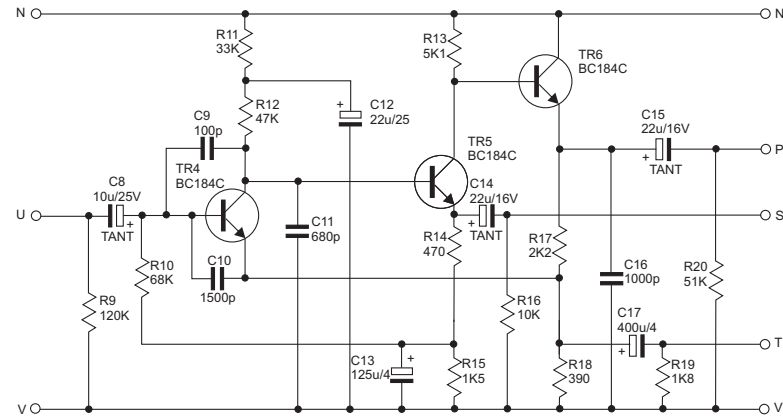




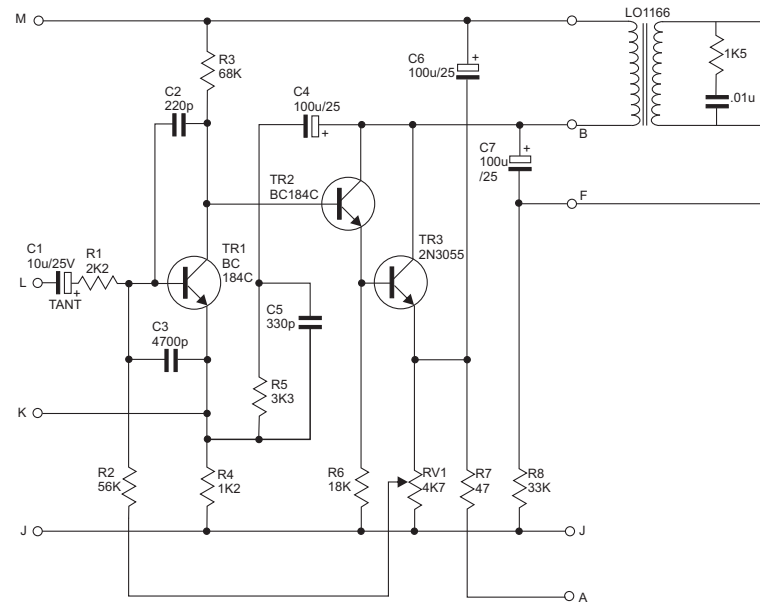


EZ1290 SCHEMATIC AND WIRING DIAGRAM

BA183NV



BA183AM



NOTE: BA183 / AV CONSISTS OF BA183AM + BA183NV

Transistor Voltages

	Collector	Base	Emitter
TR1 BC184C	3.6	0.9	0.4
TR2 BC184C	22.6	3.6	3.1
TR3 2N3055	22.6	3.1	2.5
TR4 BC184C	3.8	2.4	1.8
TR5 BC184C	12.4	3.8	3.2
TR6 BC184C	20.6	12.4	11.8

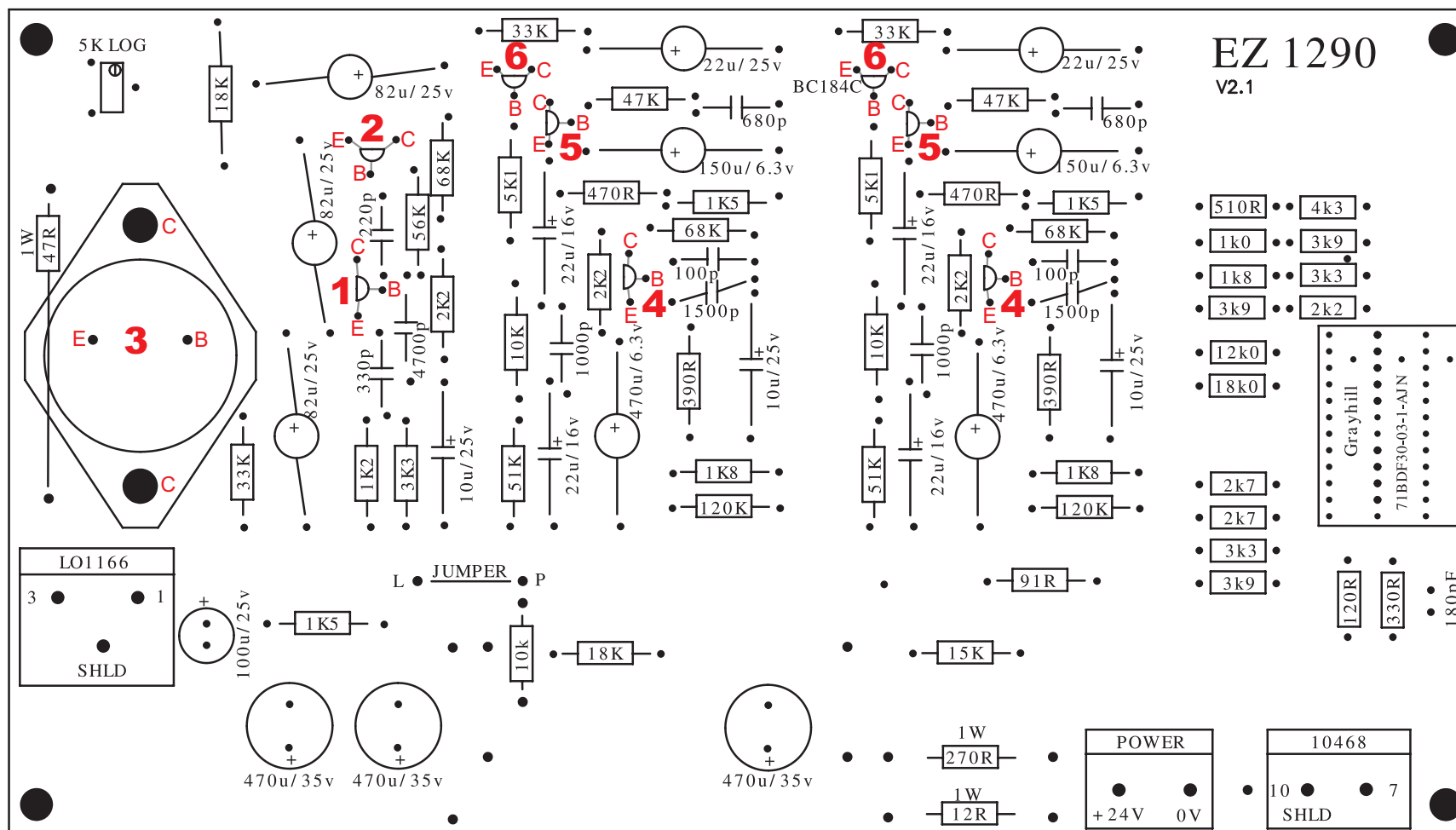
Notes:

Voltages were measured with the rail adjusted to +24.05V

Actual voltages may vary from unit to unit, BUT:

Collector should always be most positive

Base voltage should always be .5 to .6V higher than Emitter



Trim Fader

- An optional trim fader may be connected as pictured:
- This fader attenuates the output of the pre stage(s) (BA183NV) by 0 to INF dB.
 - Note that attenuation occurs before the driver stage (BA183AM).
- Be sure to use a very high quality, conductive plastic potentiometer.
 - Potentiometers by Penny and Giles, Alps are worth considering.
 - It is probably best to use no trim fader at all.

